

Manual Change History (4021499) K19 Service Manual

Date	19-Oct-2004
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	18-002-004-tr (/qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated Vacuum Test section.
Date	11-Nov-2010
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	18-012-014-tr (/qs3/pubsys2/xml/en/procedures/18/18-012-014-tr.html)
Procedure Title	Air Compressor
Update Reason	Updated procedure to include new mounting bracket.
Date	10-Dec-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-054-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Updated procedure to include steps for offset pin pistons.
Date	07-Jan-2011
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	18-013-005-tr (/qs3/pubsys2/xml/en/procedures/18/18-013-005-tr.html)
Procedure Title	Drive Belt, Alternator
Update Reason	Added information for captive fasteners on alternator belt guard.

Date	07-Jan-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	18-008-001-tr (/qs3/pubsys2/xml/en/procedures/18/18-008-001-tr.html)
Procedure Title	Belt Guard
Update Reason	Added to Section 8.
Date	04-Apr-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-043 (/qs3/pubsys2/xml/en/procedures/18/18-001-043.html)
Procedure Title	Piston
Update Reason	Updated requirements for inspect for reuse.
Date	28-Sep-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	18-008-087-tr (/qs3/pubsys2/xml/en/procedures/18/18-008-087-tr.html)
Procedure Title	Cooling Fan Belt Tensioner
Update Reason	Addition of procedure to outline and manual.
Date	19-Dec-2011
Section	Section 3 - Rocker Levers
Procedure Number	18-003-006-tr (/qs3/pubsys2/xml/en/procedures/18/18-003-006-tr.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Added: Use service Part Number 3163171 (Intake) or Part Number 3163172 (exhaust).
Date	19-Dec-2011
Section	Section 3 - Rocker Levers
Procedure Number	18-003-007-tr (/qs3/pubsys2/xml/en/procedures/18/18-003-007-tr.html)
Procedure Title	Overhead Set (Dial Indicator)
Update Reason	Added: Use service Part Number 3163171 (Intake) or Part Number 3163172 (exhaust).

Date	02-May-2012
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	18-002-004-tr (/qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Added new valve spring technical information.
Date	30-Apr-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-046-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-046-tr.html)
Procedure Title	Piston Cooling Nozzle
Update Reason	Added instructions for two piece nozzle design.
Date	27-Jun-2012
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	18-006-025 (/qs3/pubsys2/xml/en/procedures/18/18-006-025.html)
Procedure Title	Static Injection Timing
Update Reason	
Date	27-Jun-2012
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	50-006-025 (/qs3/pubsys2/xml/en/procedures/50/50-006-025.html)
Procedure Title	Static Injection Timing
Update Reason	Changed Control Part List (CPL) Manual, Bulletin to 4021328, and added refer to Procedure 99-850-013.

Date	25-Jun-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-028-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Updated the Remove step to add instruction that liner seals must be discarded if removed from the liner.
Date	20-Aug-2012
Section	Section TS - Troubleshooting System
Procedure Number	4021499-t102-tr (/qs3/pubsys2/xml/en/manual/4021499/4021499-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Changed first step to include inspection of OEM components for external oil leaks.
Date	28-June-2013
Section	Section 3 - Rocker Levers
Procedure Number	18-003-013-tr (/qs3/pubsys2/xml/en/procedures/18/18-003-013-tr.html)
Procedure Title	Rocker Lever Housing
Update Reason	Install Step: Added statement, checking for capscrews for damage. Change capscrew torque to 122 N.m [90 ft-lb]. Change graphic.
Date	05-Nov-2013
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	102-016-008 (/qs3/pubsys2/xml/en/procedures/102/102-016-008.html)
Procedure Title	Flywheel Ring Gear
Update Reason	Updated procedure for improved clarity and safety.

Date	14-Jan-2014
Section	Section TS - Troubleshooting System
Procedure Number	4021499-t102-tr (/qs3/pubsys2/xml/en/manual/4021499/4021499-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Added step three to TS tree. Verifying original equipment manufacturer (OEM) lubricating oil system components.
Date	08-Oct-2014
Section	Section TS - Troubleshooting Symptoms - Group t00
Procedure Number	4021499-t107-tr (/qs3/pubsys2/xml/en/manual/4021499/4021499-t107-tr.html)
Procedure Title	Lubricating Oil Temperature Above Specification
Update Reason	Updated TS tree with correct wording for field service on lubricating oil above or below specification.
Date	04-Dec-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-026 (/qs3/pubsys2/xml/en/procedures/18/18-001-026.html)
Procedure Title	Cylinder Block
Update Reason	Updated the Clean and Inspect for Reuse step to add a note and specifications for the main bearing bore inside diameter measurement.
Date	27-Jan-2015
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	18-003-004-tr (/qs3/pubsys2/xml/en/procedures/18/18-003-004-tr.html)
Procedure Title	Overhead Set
Update Reason	Update to service manual outline. Aligns outline with KTA19 SLB outline.

Date	07-Apr-2015
Section	Section 8 - Cooling System - Group 08
Procedure Number	18-008-018-tr (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html)
Procedure Title	Cooling System
Update Reason	Added direction to allow coolant to cool before draining.

Date	06-Apr-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-058 (/qs3/pubsys2/xml/en/procedures/18/18-001-058.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Added procedure to manual.

Date	09-Jul-2015
Section	Section 8 - Cooling System - Group 08
Procedure Number	18-008-062-tr (/qs3/pubsys2/xml/en/procedures/18/18-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Added and updated the Disassemble, Clean and Inspect for Reuse, and Assemble steps.

Date	03-Aug-2015
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-027-tr (/qs3/pubsys2/xml/en/procedures/18/18-007-027-tr.html)
Procedure Title	Lubricating Oil Pan Adapter
Update Reason	Added direction on the proper instalation of the RTV sealant and torque sequence.

Date	13-Aug-2015
Section	Section 1 - Cylinder Block
Procedure Number	18-001-028-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Updated to new geometry for standard cylinder liner sizes.
Date	20-Aug-2015
Section	Section 9 - Drive units - Group 09
Procedure Number	18-009-016-tr (/qs3/pubsys2/xml/en/procedures/18/18-009-016-tr.html)
Procedure Title	Hydraulic Pump Drive
Update Reason	Corrected inconsistencies in verbiage used and torque specifications.
Date	27-Oct-2015
Section	Section 14 - Engine Testing - Group 14
Procedure Number	18-014-008 (/qs3/pubsys2/xml/en/procedures/18/18-014-008.html)
Procedure Title	Engine Testing (In Chassis)
Update Reason	Procedure has been commonized for all high horsepower engine families.
Date	07-Jun-2016
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-007 (/qs3/pubsys2/xml/en/procedures/18/18-007-007.html)
Procedure Title	Lubricating Oil Cooler Element
Update Reason	Changes have been made to the cleaning recommendations for oil cooler elements.

Date	20-Sep-2016
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	18-003-006-tr (/qs3/pubsys2/xml/en/procedures/18/18-003-006-tr.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Updated the overhead adjustment kit part number.
Date	27-Mar-2017
Section	Section 8 - Cooling System - Group 08
Procedure Number	18-008-002-tr (/qs3/pubsys2/xml/en/procedures/18/18-008-002-tr.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	Updated to include an inspect and re-use guideline for the drive belt tensioner.
Date	04-Apr-2017
Section	Section 1 - Cylinder Block
Procedure Number	18-001-006-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-006-tr.html)
Procedure Title	Bearings, Main
Update Reason	Updated main bearing specification to match the current product.
Date	13-Sep-2017
Section	Section 1 - Cylinder Block
Procedure Number	18-001-029-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-029-tr.html)
Procedure Title	Gear Cover Spacer Plate
Update Reason	Updated the Install Step for new gasket.

Date	24-Oct-2017
Section	Section 1 - Cylinder Block
Procedure Number	18-001-036-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-036-tr.html)
Procedure Title	Idler Gear, Camshaft
Update Reason	Updated component inspection instructions before reuse and change the end clearance measurement on the camshaft idler gear to reflect the correct measurement.
Date	10-Nov-2017
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-017 (/qs3/pubsys2/xml/en/procedures/18/18-007-017.html)
Procedure Title	Lubricating Oil Filter Head (Remote-Mounted)
Update Reason	Added a note in the Install step.
Date	10-Nov-2017
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-043 (/qs3/pubsys2/xml/en/procedures/18/18-001-043.html)
Procedure Title	Piston
Update Reason	Updated instructions to include the correct information for the piston pin diameter and length and the associated piston pin bore diameter tolerances.
Date	26-Jan-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-035 (/qs3/pubsys2/xml/en/procedures/18/18-007-035.html)
Procedure Title	Lubricating Oil Suction Tube (Block-Mounted)
Update Reason	Added to manual.

Date	29-Jan-2018
Section	Section 9 - Drive Units - Group 09
Procedure Number	18-009-011-tr (/qs3/pubsys2/xml/en/procedures/18/18-009-011-tr.html)
Procedure Title	Fuel Pump Drive
Update Reason	Word change in Assemble step: lengths to diameters when determining capscrew torque.

Date	19-Jul-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-050 (/qs3/pubsys2/xml/en/procedures/18/18-007-050.html)
Procedure Title	Cylinder Smokes (Blue Smoke)
Update Reason	Added direction for the manifold to cool before removing.

Date	22-Oct-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-016-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Revised in include instructions on how to inspect removable counterweights, if equipped.

Date	22-Oct-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-015-tr (/qs3/pubsys2/xml/en/procedures/18/18-007-015-tr.html)
Procedure Title	Lubricating Oil Filter Head
Update Reason	Revised the free length and working height of the bypass and piston cooling springs.

Date	21-Nov-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-050 (/qs3/pubsys2/xml/en/procedures/18/18-007-050.html)
Procedure Title	Cylinder Smokes (Blue Smoke)
Update Reason	Procedure was updated to make it clear not to run the engine with the exhaust manual OFF.
Date	02-Jan-2019
Section	Section 1 - Cylinder Block
Procedure Number	18-001-026 (/qs3/pubsys2/xml/en/procedures/18/18-001-026.html)
Procedure Title	Cylinder Block
Update Reason	Revised, Counterbore Machining Specifications table.
Date	02-Jan-2019
Section	Section 1 - Cylinder Block
Procedure Number	18-001-028 (/qs3/pubsys2/xml/en/procedures/18/18-001-028.html)
Procedure Title	Cylinder Liner
Update Reason	Revised, Mechanically Actuated Injectors table.
Date	23-May-2019
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	18-003-001-tr (/qs3/pubsys2/xml/en/procedures/18/18-003-001-tr.html)
Procedure Title	Crankcase Breather (External)
Update Reason	New plastic breather option added to Rail Application.

Date	30-Jul-2019
Section	Section 0 - Product - Group 00
Procedure Number	18-022-001_00 (/qs3/pubsys2/xml/en/procedures/18/18-022-001_00.html)
Procedure Title	Service Tools
Update Reason	Updated the part number for the 100- to 600 ft-lb torque wrench.
Date	09-Oct-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-014 (/qs3/pubsys2/xml/en/procedures/18/18-001-014.html)
Procedure Title	Connecting Rod
Update Reason	Corrected the inside diameter of the large end of the connecting rod.
Date	15-Oct-2019
Section	Section 5 - Fuel System - Group 05
Procedure Number	18-005-007 (/qs3/pubsys2/xml/en/procedures/18/18-005-007.html)
Procedure Title	EFC Actuator Valve
Update Reason	Revised torque values.
Date	03-Feb-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	18-022-001_14 (/qs3/pubsys2/xml/en/procedures/18/18-022-001_14.html)
Procedure Title	Service Tools
Update Reason	Added cylinder diagnostic tool to Service Tools table.

Date	06-Feb-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-029-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-029-tr.html)
Procedure Title	Gear Cover Spacer Plate
Update Reason	Added note to use the Loctite™ 577, Part Number 4919202, or equivalent to install new studs.
Date	20-Mar-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-016-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Corrected listed specifications for the Rod Bearing Journal Outside Diameter, Main Bearing Journal Outside Diameter, and Thrust Distance.
Date	25-Mar-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	00-014-012 (/qs3/pubsys2/xml/en/procedures/00/00-014-012.html)
Procedure Title	Engine Compression
Update Reason	Added procedure to manual.
Date	26-Mar-2020
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	18-012-020 (/qs3/pubsys2/xml/en/procedures/18/18-012-020.html)
Procedure Title	Air Compressor (Oil Carryover)
Update Reason	Reconfigured procedure to commonize it.

Date	10-Jun-2021
Section	Section 10 - Air Intake System - Group 10
Procedure Number	18-010-040 (/qs3/pubsys2/xml/en/procedures/18/18-010-040.html)
Procedure Title	Turbocharger Compressor Seal Leaks
Update Reason	Added to manual.

Date	10-Jun-2021
Section	Section 10 - Air Intake System - Group 10
Procedure Number	18-010-049 (/qs3/pubsys2/xml/en/procedures/18/18-010-049.html)
Procedure Title	Turbocharger Turbine Seal Leaks
Update Reason	Added to manual.

Date	10-Jun-2021
Section	Section 10 - Air Intake System - Group 10
Procedure Number	18-010-033-tr (/qs3/pubsys2/xml/en/procedures/18/18-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Added instruction to check for any oil leaks on the turbine or compressor seals. Added reference to procedures 010-040 and 010-049.

Date	10-Jun-2021
Section	Section 10 - Air Intake System - Group 10
Procedure Number	18-010-037-tr (/qs3/pubsys2/xml/en/procedures/18/18-010-037-tr.html)
Procedure Title	Turbocharger, Water-Cooled
Update Reason	Added instruction to check for any oil leaks on the turbine or compressor seals.

Date	10-Jun-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-033-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-033-tr.html)
Procedure Title	Gear Housing, Front
Update Reason	Added instruction to remove the gear cover spacer plate in Preparatory Steps. Added instruction to install the gear cover spacer plate in Finishing Steps.
Date	29-Jun-2021
Section	Section 4 - Cam Followers/Tappets - Group 04
Procedure Number	18-004-014-tr (/qs3/pubsys2/xml/en/procedures/18/18-004-014-tr.html)
Procedure Title	Push Rods or Tubes
Update Reason	Minor wording correction made to procedure.
Date	29-Jun-2021
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	18-002-004-tr (/qs3/pubsys2/xml/en/procedures/18/18-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Corrected Vaccum Test step.
Date	25-Aug-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-054-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Added instruction to remove and install water bypass tube.

Date	25-Aug-2021
Section	Section 8 - Cooling System - Group 08
Procedure Number	18-008-013 (/qs3/pubsys2/xml/en/procedures/18/18-008-013.html)
Procedure Title	Coolant Thermostat
Update Reason	Added instruction to remove seal on the air compressor coolant return tube. Added instruction to install seal on the bypass tube.

Date	25-Aug-2021
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021499-t122-tr (/qs3/pubsys2/xml/en/manual/4021499/4021499-t122-tr.html)
Procedure Title	Turbocharger Leaks Engine Oil or Fuel
Update Reason	Added references to procedures 010-040 and 010-049.

Date	20-Sep-2022
Section	Section 8 - Cooling System - Group 08
Procedure Number	18-008-036-tr (/qs3/pubsys2/xml/en/procedures/18/18-008-036-tr.html)
Procedure Title	Fan Hub, Belt Driven
Update Reason	Noted that the type of grease utilized for lubrication must meet the NGLI GC-LG wheel bearing grease requirements.

Date	02-Jan-2023
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	18-006-003 (/qs3/pubsys2/xml/en/procedures/18/18-006-003.html)
Procedure Title	Air in Fuel
Update Reason	Deleted and added text to revise actions to take if an air leak is found.

Date	03-Jan-2023
Section	Section V - Specifications
Procedure Number	18-018-017 (/qs3/pubsys2/xml/en/procedures/18/18-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Changed oil specification chart to reflect proper values.

Date	27-Jan-2023
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	18-007-037-tr (/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html)
Procedure Title	Lubricating Oil System
Update Reason	Added the torque value for straight thread o-ring oil pan drain plugs to the procedure.

Date	10-Mar-2023
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	18-001-036-tr (/qs3/pubsys2/xml/en/procedures/18/18-001-036-tr.html)
Procedure Title	Idler Gear, Camshaft
Update Reason	Removed reference to procedure 001-999. Moved information for press-fit idler shaft installation above the Finishing Steps of procedure.

Last Modified: 14-Mar-2023
